

Fetal Health Multiclassification Using Machine Learning

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UE24CS352A — Machine Learning

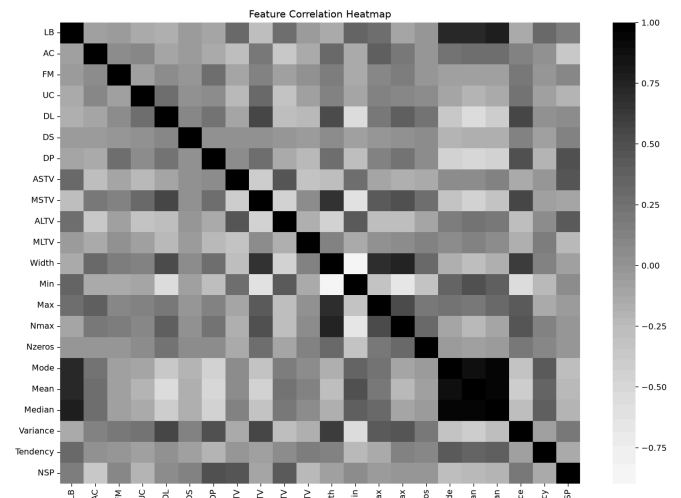
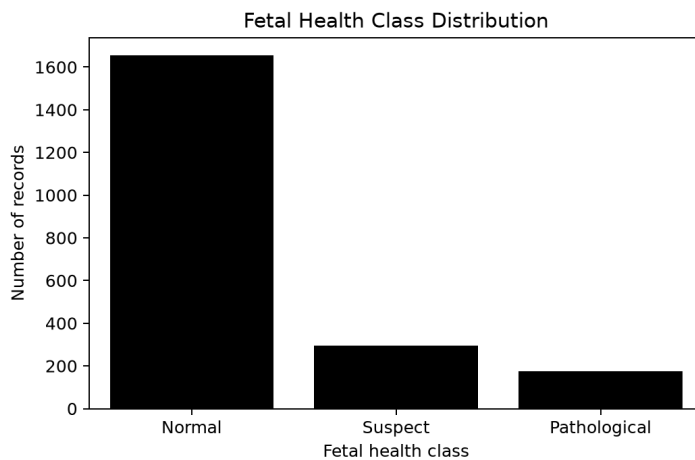
1. Problem Statement

Cardiotocography (CTG) records contain fetal heart-rate, movement, uterine-contraction and variability measurements. The objective is to classify fetal health into three NSP classes: **Normal**, **Suspect** and **Pathological** using supervised machine learning and compare multiple classification models.

2. Dataset

The project uses the **UCI Cardiotocography dataset**, containing **2,126 records** and **21 predictive features**. The target variable is NSP, with three classes: Normal, Suspect and Pathological. The dataset is imbalanced, so **macro-averaged metrics** are used alongside accuracy and weighted F1.

Records	Features	Target	Classes
2,126	21 CTG features	NSP	Normal / Suspect / Pathological



3. Approach and Methodology

A stratified 70:30 train-test split with `random_state=42` was used. `StandardScaler` was fitted only on training data and applied to scale-sensitive models, while tree-based models used the original features. Ten models were evaluated: Logistic Regression, Linear SVM, KNN, Decision Tree, Random Forest, Gradient Boosting, SVM, MLP, XGBoost and LightGBM. Performance was measured using accuracy, macro precision, macro recall, macro F1 and weighted F1.

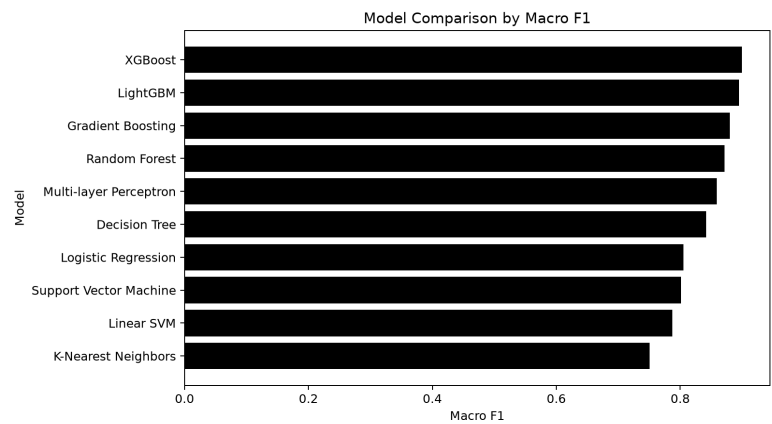
4. Implementation

The project is organized into Python modules for preprocessing, model definition, evaluation and experiment execution. The pipeline automatically generates evaluation metrics, confusion matrices and result plots. A **Random Forest feature-importance analysis** was added for interpretability. The top three features were **ASTV**, **MSTV** and **ALTV**.

5. Experimental Results

The models were compared on the held-out test set. XGBoost achieved the best overall performance, with 94.36% accuracy and 89.98% macro F1.

Model	Accuracy	Macro F1
XGBoost	94.36%	89.98%
LightGBM	94.20%	89.49%
Gradient Boosting	93.42%	88.04%
Random Forest	92.95%	87.15%
Multi-layer Perceptron	92.79%	85.89%
Decision Tree	90.75%	84.19%
Logistic Regression	89.34%	80.52%
Support Vector Machine	90.28%	80.14%
Linear SVM	88.56%	78.76%
K-Nearest Neighbors	87.62%	75.05%



Best observed result: XGBoost — 94.36% accuracy and 89.98% macro F1.

6. Feature Analysis

Random Forest feature-importance analysis ranked the CTG variables. The top three features in the current run were ASTV (abnormal short-term variability), MSTV (mean short-term variability) and ALTV (percentage of time with abnormal long-term variability). The analysis produces both `feature_importance.csv` and `feature_importance.png` in the `results/` directory.

Rank	Feature	Importance
1	ASTV	0.136098
2	MSTV	0.122950
3	ALTV	0.114367

7. Conclusion and Limitations

The experiments show that **boosting and ensemble models performed strongly**, with XGBoost achieving the highest accuracy and macro F1. Feature analysis also highlighted fetal heart-rate variability measures as important predictors. This is an **academic ML prototype**, not a clinically validated diagnostic system. Limitations include class imbalance, a single train-test split and lack of clinical validation.